

ENERGY-EFFICIENT WIRELESS COMMUNICATION STRATEGY FOR SENSOR NODES

Design Task – CO3 | Advance Wireless Communication Systems / 5G / 6G Communication Systems

Course Code: ECA56 Assessment: CO3-AT2 Design Task Marks: 100

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1. Problem Statement

Wireless sensor nodes are generally battery-powered and may operate for long periods without physical access. Major energy losses occur during wireless transmission, long-distance communication, interference, retransmissions, idle listening, and unnecessary signalling. This design proposes an energy-efficient 5G/6G-oriented wireless communication strategy that maximizes network lifetime while maintaining reliable data transmission and efficient spectrum utilization.

2. Proposed System Architecture

Sensor Nodes → Cluster/Gateway → 5G/6G Base Station → Edge/Cloud Server

Each sensor node contains a sensing unit, microcontroller, energy-efficient transceiver, battery or energy-harvesting unit, and wireless communication module. The gateway collects and aggregates sensor data, performs resource allocation and power-control decisions, and communicates with the 5G/6G base station. The base station provides OFDMA resource allocation, NOMA, Massive MIMO beamforming, dynamic spectrum access, and adaptive transmission.

3. Proposed Energy-Efficient Strategy

The design combines dynamic spectrum access, OFDMA resource-block allocation, power-domain NOMA with SIC, adaptive transmission-power control, Massive MIMO beamforming, sleep/wake scheduling, and data aggregation.

4. Dynamic Spectrum Access

A cognitive-radio-enabled sensor network senses available channels before transmission. Instead of remaining on a congested fixed channel, the node selects an available channel with good channel quality and low interference. A channel-selection metric can be expressed as:

$$C_{\text{best}} = \arg \max_C [(\text{Available Bandwidth} \times \text{Channel Quality}) / (\text{Interference} + \text{Noise})]$$

This reduces congestion, interference, and retransmissions while improving spectrum utilization.

5. OFDMA Resource Block Allocation

OFDMA divides the available bandwidth into multiple resource blocks (RBs). The gateway assigns RBs according to channel quality, battery level, data priority, distance, and required data rate.

Sensor	Requirement	Assigned RB	Priority
S1	Temperature	RB1	Low
S2	Pressure	RB2	Medium
S3	Emergency	RB3	Very High
S4	Humidity	RB4	Low

An energy-aware allocation metric can be defined as $M_i = R_i / P_i$, where R_i is achievable data rate and P_i is transmission power.

6. NOMA and Successive Interference Cancellation

Power-domain NOMA allows multiple compatible sensor nodes to share the same resource block. The transmitted signal can be represented as:

$$x = \sqrt{P_1}x_1 + \sqrt{P_2}x_2$$

Users with weaker channels can be assigned higher power, while stronger users can use lower power. At the receiver, Successive Interference Cancellation (SIC) separates the superimposed signals. Sharing the same resource reduces waiting time, spectrum usage, and signalling overhead.

7. Adaptive Transmission Power Control

Sensor nodes should not always transmit at maximum power. Required power is adapted according to distance, path loss, channel quality, interference, and required data rate. A simplified model is:

$$P_t = P_{\min} (d/d_0)^n$$

where P_t is transmission power, d is sensor-to-gateway distance, d_0 is reference distance, and n is the path-loss exponent. Nearby sensors can therefore operate with lower power while maintaining reliable communication.

8. Massive MIMO and Beamforming

A 5G/6G base station equipped with Massive MIMO can form directional beams toward sensor nodes. Beamforming improves received signal strength, reduces interference, increases coverage, and improves spectral efficiency. As a result, sensor nodes can transmit at lower power for the same reliability target.

9. Sleep/Wake Scheduling

Duty cycling is used to prevent energy loss during idle periods. Each sensor operates in three main states: Sleep, Wake/Receive, and Transmit. If no new measurement or transmission request exists, the node enters sleep mode. It wakes only when sensing or communication is required. This significantly reduces idle-listening energy.

10. Data Aggregation

The gateway aggregates correlated measurements before forwarding them to the base station. For example, the average temperature can be calculated as:

$$T_{\text{avg}} = (T_1 + T_2 + \dots + T_n) / n$$

Instead of transmitting every individual measurement, the gateway can transmit the required aggregated information. This reduces packet transmissions, channel occupation, traffic, and energy consumption.

11. Complete Communication Algorithm

1. Sensor node measures the physical parameter.
2. Check whether new data is available.
3. If no data is available, enter sleep mode.
4. If data is available, evaluate battery level, channel quality, distance, and required data rate.
5. Perform spectrum sensing and identify available channels.
6. Select the best available channel using dynamic spectrum access.
7. Gateway assigns an OFDMA resource block.
8. If compatible users are available, apply power-domain NOMA.

9. Determine the minimum required transmission power.
10. Transmit using the selected resource and adaptive modulation/rate.
11. Base station applies Massive MIMO beamforming.
12. Perform SIC when NOMA is used.
13. Aggregate correlated data at the gateway.
14. Return the sensor node to sleep mode when communication is complete.

12. Energy Consumption Model

$$E_{\text{total}} = E_{\text{sense}} + E_{\text{process}} + E_{\text{tx}} + E_{\text{rx}}$$

where E_{sense} is sensing energy, E_{process} is processing energy, E_{tx} is transmission energy, and E_{rx} is reception energy. Transmission energy is:

$$E_{\text{tx}} = P_{\text{tx}} \cdot T_{\text{tx}}$$

Therefore, reducing transmission power and/or transmission time directly reduces energy consumption.

13. Energy-Efficiency Contributions

Technique	Energy-saving mechanism
Dynamic Spectrum Access	Reduces interference and retransmissions
OFDMA	Efficient resource-block allocation
NOMA	Allows multiple users to share a resource
SIC	Separates simultaneously transmitted signals
Power Control	Avoids unnecessary transmit power
Massive MIMO	Beamforming improves link quality
Sleep/Wake	Eliminates idle listening
Data Aggregation	Reduces number of transmitted packets
Adaptive Rate/Modulation	Avoids unnecessary high-rate transmission

14. Performance Evaluation Parameters

Energy Efficiency: $EE = \text{Throughput} / \text{Energy Consumption}$, measured in bits/Joule.

Network Lifetime: Time until the defined percentage of sensor nodes exhaust their batteries.

Packet Delivery Ratio: $PDR = (\text{Successfully Received Packets} / \text{Transmitted Packets}) \times 100$.

Spectral Efficiency: $SE = \text{Data Rate} / \text{Bandwidth}$, measured in bits/s/Hz.

End-to-End Delay: Time required for sensor information to reach the destination.

15. Expected Results

The proposed strategy is expected to achieve lower energy consumption, longer sensor battery lifetime, higher spectral efficiency, reduced transmission delay, lower interference, improved packet delivery ratio, better coverage, and reliable support for dense IoT deployments.

16. Comparison with Conventional Sensor Communication

Parameter	Conventional Approach	Proposed Approach
Spectrum	Fixed channel	Dynamic spectrum access

Parameter	Conventional Approach	Proposed Approach
Multiple access	Separate transmissions	OFDMA + NOMA
Power	Fixed/high power	Adaptive power control
Antennas	Conventional antenna	Massive MIMO beamforming
Idle operation	Continuous listening	Sleep/wake scheduling
Data transmission	Individual packets	Aggregation where applicable
Expected energy	Higher	Lower
Network lifetime	Lower	Higher

17. Conclusion

The proposed energy-efficient wireless communication strategy integrates dynamic spectrum access, OFDMA, NOMA with SIC, adaptive power control, Massive MIMO beamforming, sleep/wake scheduling, and data aggregation for battery-powered IoT sensor nodes. Dynamic spectrum access improves spectrum utilization, OFDMA provides flexible resource-block allocation, NOMA increases simultaneous access, and adaptive power control reduces unnecessary transmission energy. Massive MIMO improves link quality through beamforming, while sleep scheduling and data aggregation further reduce energy usage. The overall design provides a practical 5G/6G-oriented solution for improving energy efficiency, spectral efficiency, reliability, and network lifetime.

18. Suggested Implementation for GitHub

Repository structure:

```
Energy-Efficient-Wireless-Sensor-Network/
■■■■ README.md
■■■■ report/design_report.pdf
■■■■ simulation/energy_efficiency_simulation.py
■■■■ results/energy_consumption.png
■■■■ results/throughput.png
■■■■ results/network_lifetime.png
■■■■ diagrams/system_architecture.png
```

Note: For a BL6 design assessment, adding a Python simulation comparing the proposed strategy with a conventional wireless sensor network will strengthen the design justification and evaluation.